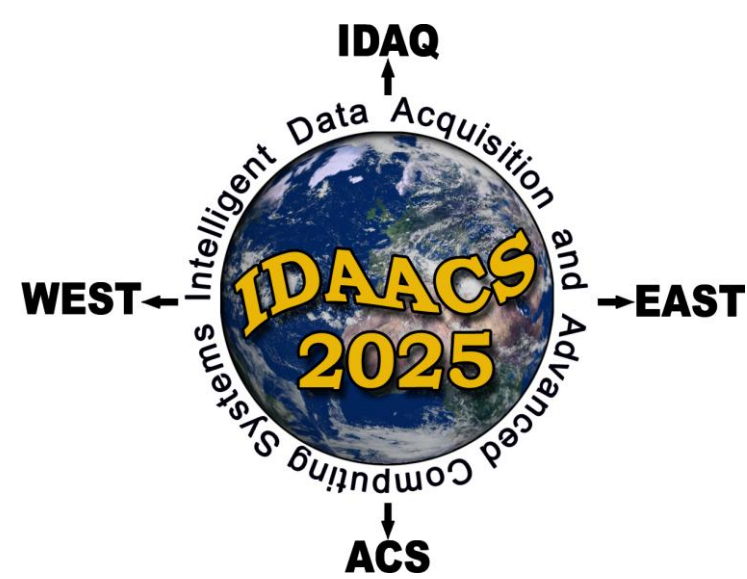




YOLO Ensemble for UAV-based Multispectral Defect Detection in Wind Turbine Components

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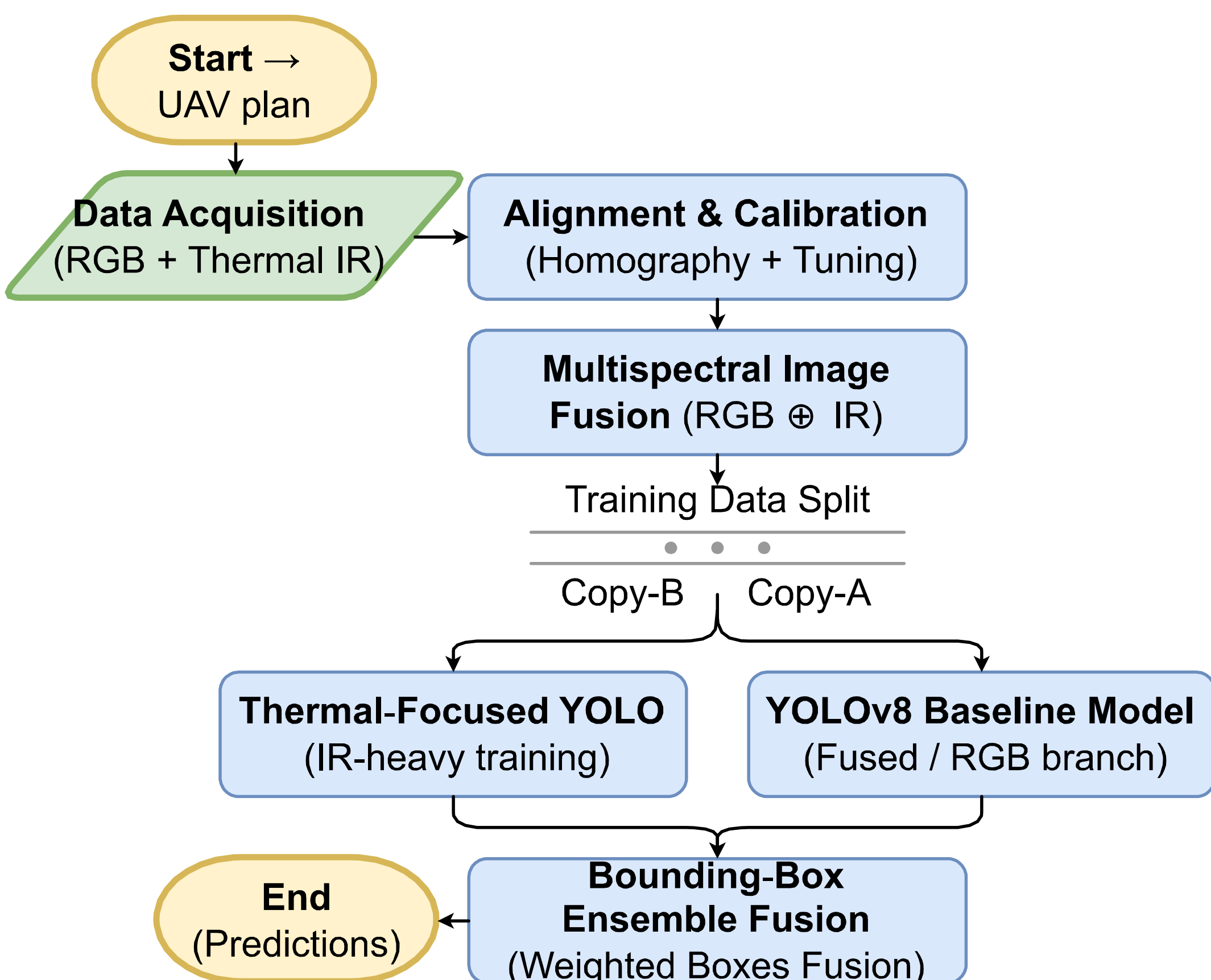
INTRODUCTION

- Wind power plants critical in renewable energy transition
- Efficiency depends on timely maintenance and defect prevention
- Minor defects can pose substantial economic and safety risks
- UAVs with sensors efficiently collect visual information

MAJOR CONTRIBUTIONS

- Multispectral image fusion (**RGB + thermal IR**)
- Novel ensemble learning approach with bounding box fusion
- Comprehensive evaluation showing superior performance

METHODS



Our approach integrates high-resolution visible (**RGB**) and thermal (**IR**) data channels through:

- Data acquisition
- Multispectral image fusion
- Parallel model training
- Ensemble fusion

Data Collection and Model Training:

- RGB Channel: High-resolution optical images (4K/6K)
- IR Channel: Thermal images for temperature anomalies
- Three defect classes: **Cracks** (C^1), **Corrosion** (C^2), **Overheating** (C^3)

ENSEMBLE FUSION ALGORITHM

Intelligent combination of predictions using **bounding box fusion**:

- Merges overlapping detections of the same class
- Preserves unique findings from each model
- Uses weighted averages and Non-Maximum Suppression

RESULTS

Training dataset: Blade30 dataset (1,102 images) + 670 overheating images

Model	mAP@.5	mAP@.5:.95	F1-Score
YOLOv8 (Single)	0.91	0.76	0.88
Faster R-CNN	0.89	0.74	-
Cascade R-CNN	0.90	0.75	-
Our Ensemble	0.93	0.79	0.90

Key improvements:

- Mean mAP@.5: 0.91 → 0.93 (2.2% improvement)
- F1-score for "Cracks": 0.90 → 0.92
- F1-score for "Overheating": 0.86 → 0.89

CONCLUSIONS

- Our ensemble approach achieves superior performance in detecting wind turbine defects compared to single-model approaches
- Integration of thermal data is particularly impactful for identifying overheating and subsurface defects
- Future research will focus on optimizing the ensemble for real-time edge deployment and incorporating advanced data fusion techniques